

Validation Report: Dynamic Modelling and Simulation of a Conveyor Belt

1. Objective of Validation

The purpose of validation is to verify that the developed dynamic model of the conveyor belt accurately represents the physical system and produces results consistent with analytical calculations and physical laws.

2. Validation 1 – Energy Consistency (CO5)

At steady state, $KE = \frac{1}{2} mv^2 = 1.125 \text{ J}$. Simulation result $\approx 1.12 \text{ J}$. The simulated value closely matches analytical computation.

3. Validation 2 – Analytical Time to Reach Belt Speed (CO4)

$t = V_{\text{belt}}/(\mu g) = 1.5/(0.35 \times 9.81) = 0.437 \text{ s}$. Simulation shows $\approx 0.43 \text{ s}$. Error $\approx 1.6\%$.

4. Validation 3 – Dimensional Consistency Check

$dv/dt = \mu g$. Units on both sides are m/s^2 , confirming dimensional correctness.

5. Sensitivity Analysis

Varying μ from 0.2 to 0.5 changes acceleration time from 0.76 s to 0.31 s. Higher μ leads to faster lock-in.

6. Model Limitations

Belt elasticity not considered. Motor torque dynamics excluded. No experimental validation. Longitudinal vibration ignored.

7. Overall Conclusion

The model is validated through analytical comparison, dimensional checks, and sensitivity analysis. Results show strong agreement.

Sensitivity Analysis Table

Coefficient of Friction (μ)	Acceleration (m/s^2)	Time (s)
0.20	1.962	0.76
0.35	3.433	0.43
0.50	4.905	0.31